

CS 473 (Spring 2025)
Homework 9 (due Apr 24 Thu 10am)

Instructions: As in previous homeworks.

Problem 9.1: Consider the following problem COLORFUL-PATH:

Input: a directed acyclic graph (dag) $G = (V, E)$ with n vertices and m edges, and vertices $s, t \in V$, and a color $c(e) \in \{1, \dots, M\}$ for each edge $e \in E$, and a number ℓ .

Output: yes iff there exists a path from s to t that encounters at least ℓ *distinct* colors.

Prove that COLORFUL-PATH is NP-hard, by giving a polynomial-time reduction from 3-SAT to COLORFUL-PATH. Remember to describe precisely your construction of an input to COLORFUL-PATH from any given input to 3-SAT (a 3CNF formula – note that we are not given a satisfying assignment itself!), and carefully prove correctness of your reduction.

(Hint: for your construction of the dag G , try something like the picture below... Describe how to assign colors to the edges along these paths.)

Solution:

Reduction from 3-SAT

Given a 3CNF formula with n variables $x_1, \dots, x_n$ and m clauses $C_1, \dots, C_m$, where each clause C_j contains exactly three literals.

Construct a DAG $G = (V, E)$, a coloring function $c : E \rightarrow \{1, \dots, m\}$, and vertices s and t such that the formula is satisfiable iff there exists a path from s to t in G that encounters at least $\ell = m$ distinct colors.

DAG Construction

- For each variable x_i , construct a gadget consisting of two paths from a vertex u_i to v_i :
 - The upper path P_i^T corresponds to assigning $x_i = \text{true}$.
 - The lower path P_i^F corresponds to assigning $x_i = \text{false}$.

These paths form a “diamond” structure.

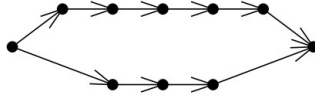


Figure 1: gadget

- Each clause C_j is assigned a unique color $j \in \{1, \dots, m\}$.
- For each occurrence of literal x_i in clause C_j , insert an edge labeled with color j on the upper path P_i^T .
- For each occurrence of literal $\neg x_i$ in clause C_j , insert an edge labeled with color j on the lower path P_i^F .
- Connect the gadgets in sequence:

$$u_1 \rightarrow v_1 \rightarrow u_2 \rightarrow v_2 \rightarrow \dots \rightarrow u_n \rightarrow v_n.$$

Merge v_i and u_{i+1} ($i = 1, 2, \dots, n-1$) into a single node u_{i+1} (no colored path between them).

Let the source vertex $s \leftarrow u_1$ and the sink vertex $t \leftarrow v_n$.

Thus we have the DAG:

$$s \rightarrow u_2 \rightarrow \dots \rightarrow u_n \rightarrow t.$$

each $\rightarrow$ represents a gadget.

Correctness of the Reduction

($\Rightarrow$) Suppose the 3CNF formula is satisfiable. Then there exists a truth assignment such that every clause C_j is satisfied. Construct an s - t path in G as follows:

- For each variable x_i , if $x_i = \text{true}$, follow the upper path P_i^T ; otherwise, follow P_i^F .

Because the assignment satisfies all clauses, for each clause C_j , at least one literal in C_j is true under the assignment. Therefore, the corresponding path will include an edge colored with j . Hence, the path encounters all m colors.

($\Leftarrow$) Suppose there is an s - t path in G that encounters at least m distinct colors. Because each gadget has only two disjoint paths, and the entire graph is a sequence of such gadgets, the path must make exactly one choice (true/false) per variable. Define the truth assignment accordingly:

- If the path takes P_i^T , assign $x_i = \text{true}$.
- If the path takes P_i^F , assign $x_i = \text{false}$.

Since the path encounters all m colors, each clause color must appear on the path. Therefore, for each clause C_j , at least one of its literals is satisfied under this assignment. Thus, the 3CNF formula is satisfiable.

Complexity

The number of vertices and edges in G is $O(nm)$. The COLORFUL-PATH problem is in NP, as the reduction can be done in polynomial time and a path can be verified in polynomial time.

Problem 9.2: Consider the following problem COVERING-SQUARE-BY-TRIANGLES:

Input: a set T of n triangles in 2D, a square s , and an integer k .

Output: yes iff there exists a subset $Q \subseteq T$ of k triangles such that the union of Q covers the entire boundary of s .

(Note: we don't require the union of Q to cover the interior of s . Also, the variant of the problem of covering just one side of s , rather than the entire boundary of s , is easier and can actually be solved in polynomial time by a greedy algorithm.)

Prove that COVERING-SQUARE-BY-TRIANGLES is NP-hard.

You may use the fact that VERTEX-COVER is NP-hard even when restricted to graphs with maximum degree 3. You may also use the fact that any graph with maximum degree d is $(d+1)$ -edge-colorable, i.e., we can assign a color from $\{1, \dots, d+1\}$ to each edge, so that no two adjacent edges have the same color; such a coloring can be found in polynomial time.

As always, describe the construction of your reduction precisely, and carefully prove correctness of your reduction.

(Hint: map each edge of the input graph to a tiny interval along the boundary of s ...)

Solution:

Lemma. Let s be a square in the plane, and on three distinct sides of s choose three disjoint open segments, **none of which touches a corner of s** . Then there exists a triangle whose edges **exactly** coincide with those three given segments.

Proof. On Side i let the open segment be $[u_i, v_i]$, oriented so that as we traverse the boundary of s in clockwise order we meet u_i before v_i . Since all three segments have the same length and lie on distinct sides, the six points

$$u_1, v_1, u_2, v_2, u_3, v_3$$

appear in convex position along the boundary of s .

Consider the three lines

$$\ell_1 = \overline{v_1 u_2}, \quad \ell_2 = \overline{v_2 u_3}, \quad \ell_3 = \overline{v_3 u_1}.$$

We claim that no two of these lines are parallel.

Suppose, for sake of contradiction, that ℓ_i and ℓ_j were parallel. Because each segment $[u_k, v_k]$ lies on a different side of the square, the only way, say, $\overline{v_1 u_2} \parallel \overline{v_2 u_3}$ can occur is if those lines both lie in directions parallel to one pair of opposite sides of the square. But then the third line $\ell_3 = \overline{v_3 u_1}$ must connect points on the remaining pair of opposite sides, and so ℓ_3 would pass through a corner of the square. This contradicts the segment construction.

Hence ℓ_1, ℓ_2, ℓ_3 are pairwise non-parallel, and so they form the three sides of a triangle T .

Reduction Construction

Idea: We reduce from VERTEX-COVER on graphs with maximum degree 3. Given a graph $G = (V, E)$ with $\Delta(G) \leq 3$ and an integer K , we construct an instance of COVERING-SQUARE-BY-TRIANGLES.

1. Since $\Delta(G) \leq 3$, the graph is 4-edge-colorable. Compute in polynomial time a proper 4-edge-coloring

$$c(e) = j, \quad j \in \{1, 2, 3, 4\}, \quad e \in E.$$

Let

$$k = \max_{j \in \{1, 2, 3, 4\}} |\{e \in E : c(e) = j\}|.$$

2. Label the four sides of a square s by 1, 2, 3, 4. On side j partition its interior into $k + 2$ equal open segments

$$S_{j,1}, S_{j,2}, \dots, S_{j,k+2}.$$

Since each color class has at most k edges, we can assign each edge $e \in E$ of color j to a distinct segment $S_{j,i}$. The remaining two segments covering vertices of the square stay unused.

3. For each vertex $v \in V$ with incident edges $e_1, \dots, e_d$ ($d \leq 3$). By the lemma, there is a unique triangle T_v whose edges exactly cover $S_{c(e_d),i}$ ($d \leq 3, i \in \{1, 2, \dots, k+2\}$). (If $d < 3$, the triangle is easier to construct by adding some dummy vertices near the center of the square.)

Collect all such $\{T_v : v \in V\}$.

4. Any segment $S_{j,i}$ not used by an edge gets its own “fill-in” triangle that exactly covers that single segment. Call the set of all these triangles F .

Let

$$T = \{T_v : v \in V\} \cup F, \quad W = K + |F|.$$

Correctness of the Reduction

($\Rightarrow$) If G has a vertex cover of size K :

Let $V' \subseteq V$ be a vertex cover of size K . Then select triangles

$$Q = \{T_v : v \in V'\} \cup F,$$

which has total size $K + |F| = W$.

For every edge $e = (u, v)$, at least one of u or v is in V' , so the corresponding δ_e is covered by the triangle T_u or T_v . All remaining parts of the square boundary are covered by triangles in F , so Q covers the entire boundary.

($\Leftarrow$) If $Q \subseteq T$ of size k covers the square boundary:

Since the fill-in triangles in F are the only ones that cover the leftover segments, all of F must be included in Q .

Thus, $Q \setminus F$ has size at most K and must consist of triangles from T_V . Let $V' = \{v \in V : T_v \in Q\}$. By the Lemma, each edge is exactly covered by some triangles, and for every edge $e \in E$, the segment Δ_e must be covered, so some T_v of v incident to e must be in Q . Therefore, V' is a vertex cover.

Complexity

Note that the size of the constructed instance is polynomial in the size of the original graph. Each edge maps to a small interval on the square boundary, and each vertex yields at most one triangle. The number of fill-in triangles is smaller than $|E|$, since they only cover the leftover segments between δ_e . Hence, the total number of triangles in T is $|V| + O(|E|)$, and all geometric constructions (edge coloring, triangle placement) can be done in polynomial time. Therefore, the reduction is polynomial-time and COVERING-SQUARE-BY-TRIANGLES is NP-hard.

Problem 9.3: As we all know, an *interval* is a set $[a, b]$ containing all real numbers between a and b . Define a *circular interval* to be either an interval $[a, b]$ (if $a \leq b$), or the union of two unbounded intervals $[a, \infty) \cup (-\infty, b]$ (if $a > b$). (In other words, we allow the real line to “wrap around”.)

Consider the following problem CIRCULAR-INTERVAL-SELECTION:

Input: N circular intervals $I_1, I_2, \dots, I_N$ with positive integer weights $w_1, \dots, w_N$.

Output: a subset $S \subseteq \{1, \dots, N\}$ maximizing total weight $\sum_{i \in S} w_i$ such that for every $i, j \in S$ ($i \neq j$), the circular intervals I_i and I_j are disjoint.

It is possible to solve the CIRCULAR-INTERVAL-SELECTION problem in $O(N^2)$ time by dynamic programming (you don’t have to give such an algorithm). In contrast, without circularity, the problem can be solved in $O(N \log N)$ time. In this question, you will provide evidence on why the circular version of the problem is more difficult.

To this end, consider the following known graph problem called MIN-WEIGHT-TRIANGLE:

Input: a directed graph $G = (V, E)$ with n vertices, where each edge (u, v) has a positive integer weight $w(u, v)$.

Output: a cycle of length 3 in G with the minimum total weight, i.e., $v_i, v_j, v_k \in V$ with $(v_i, v_j), (v_j, v_k), (v_k, v_i) \in E$ minimizing $w(v_i, v_j) + w(v_j, v_k) + w(v_k, v_i)$.

It has been hypothesized that no algorithm can solve MIN-WEIGHT-TRIANGLE in $O(n^{2.999})$ time (in other words, the obvious cubic-time, brute-force algorithm is close to be the best possible).

Prove that under this hypothesis, no algorithm can solve CIRCULAR-INTERVAL-SELECTION in $O(N^{1.499})$ time.

(Hint: map each edge (v_i, v_j) of the input graph to three circular intervals $[i, M + j - 1]$, $[M + i, 2M + j - 1]$, and $[2M + i, \infty) \cup (-\infty, j - 1]$ for some M . But how to assign weights to these circular intervals?)

Solution:**Definition**

A *perfect loop* is a set of three disjoint circular intervals on the real line that join end-to-end in a cycle.

$$[k+1, \infty) \cup (-\infty, i], [i+1, j], [j+1, k], \quad i, j, k \in \mathbb{R}^+$$

We will show a bijection between directed triangles in the graph and such perfect loops in the interval instance.

Interval Construction

Let $M = n + 1$. For each directed edge $(i \rightarrow j) \in E$, construct three circular intervals:

$$\begin{aligned} I_{ij}^{(1)} &= [i, M + j - 1], \\ I_{ij}^{(2)} &= [M + i, 2M + j - 1], \\ I_{ij}^{(3)} &= [2M + i, \infty) \cup (-\infty, j - 1]. \end{aligned}$$

By construction, for any triangle $(i \rightarrow j \rightarrow k \rightarrow i)$, the three intervals

$$I_{ij}^{(1)}, \quad I_{jk}^{(2)}, \quad I_{ki}^{(3)}$$

are exactly a perfect loop as defined above. (Similarly for $I_{ij}^{(2)}, I_{jk}^{(3)}, I_{ki}^{(1)}$ and $I_{ij}^{(3)}, I_{jk}^{(1)}, I_{ki}^{(2)}$.)

Since $M > n$, any two intervals of the same type (i.e., both of the form $I^{(1)}$, $I^{(2)}$, or $I^{(3)}$) will necessarily overlap. Therefore, any disjoint subset of intervals can contain at most one interval of each type, implying that the maximum size of a disjoint subset is 3, and any such subset of size 3 must consist of exactly one interval of each type.

Weight Assignment

Let $W = 3 \cdot \max_{(i \rightarrow j) \in E} w_{ij}$. Assign

$$W_{ij}^* = (M + j - i)W - w_{ij}.$$

Since $w_{ij} > 0$ and $M = n + 1$, we have $W_{ij}^* > 0$.

For any disjoint set of size 3, its total weight is

$$(3M + v' - v + u' - u + w' - w)W - \text{some weights},$$

where

$$w' < v < v' < u < u' < w, \quad u, u', v, v', w, w' \in \mathbb{R}^+.$$

Notice that $v' - v + u' - u + w' - w \leq 0$. From this, we can conclude that for any set of three disjoint circular intervals the total weight is always at most $3MW$.

For a perfect loop, since it corresponds to a triangle, we have $v' - v + u' - u + w' - w = 0$, its total weight is at least $(3M - 1)W$. Meanwhile, for any set of three disjoint intervals that does not form a perfect loop, we have $v' - v + u' - u + w' - w < 0$, its total weight strictly less than $(3M - 1)W$.

Correctness of the Reduction

($\Rightarrow$) If $(i \rightarrow j \rightarrow k \rightarrow i)$ is a directed triangle, then selecting $I_{ij}^{(1)}, I_{jk}^{(2)}, I_{ki}^{(3)}$ yields three disjoint intervals forming a perfect loop. Their total weight is

$$3MW - (w_{ij} + w_{jk} + w_{ki}).$$

where $(w_{ij} + w_{jk} + w_{ki})$ is the weight of the triangle. Since any set of three disjoint intervals that does not form a perfect loop must have smaller weight, the MIN-WEIGHT-TRIANGLE corresponds to the max-weight CIRCULAR-INTERVAL-SELECTION.

($\Leftarrow$) Conversely, any set of three disjoint intervals that does not form a perfect circular loop will either contain overlapping intervals (and thus be invalid) or have a total weight strictly less than that of any perfect loop. Therefore, the disjoint subset with maximum total weight must consist of exactly three intervals that form a perfect circular loop. By the interval construction, such a perfect loop corresponds precisely to a directed triangle $(u \rightarrow v \rightarrow w \rightarrow u)$ in the graph G .

Complexity

This reduction runs in $O(|E|)$ time and produces $N = 3|E| = O(n^2)$ intervals. An $O(N^{1.499})$ algorithm for CIRCULAR-INTERVAL-SELECTION would yield an $O(n^{2.999})$ algorithm for MIN-WEIGHT-TRIANGLE, contradicting the conjectured hardness.